

NALCHIK, Russian Federation

WMO: 372120

Lat: 43.513N

Lon: 43.637E

Elev: 445

StdP: 96.09

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-12.6	-18.2	0.8	-16.2	-14.8	1.1	-12.1	5.2	2.0	4.2	2.2	0.6	250	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	32.9	21.3	31.1	20.8	29.9	20.4	22.9	29.7	22.0	28.9	21.2	27.8	1.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.4	27.3	19.9	15.4	25.8	19.0	14.5	24.7	70.1	29.7	66.5	29.1	63.6	27.8	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.3	3.5	-18.4	35.0	4.8	1.4	-21.8	36.0	-24.6	36.8	-27.3	37.6	-30.7	38.6		
(5)			-18.6	24.1	4.6	1.4	-21.9	25.2	-24.5	26.0	-27.1	26.8	-30.5	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-1.9	-1.1	4.6	10.4	16.5	20.8	23.2	23.2	17.9	11.3	4.7	0.9
	DBStd	9.82	5.22	5.61	3.93	4.30	3.39	2.91	2.46	2.84	3.68	4.18	4.31	3.99
	HDD10.0	1385	368	311	173	46	2	1	0	0	1	34	167	283
	HDD18.3	3141	627	543	426	238	75	9	1	2	49	220	410	541
	CDD10.0	1732	0	1	6	59	203	326	408	409	238	74	7	0
	CDD18.3	447	0	0	0	1	18	85	151	154	37	2	0	0
	CDH23.3	3886	0	0	1	21	163	692	1287	1409	296	16	0	0
	CDH26.7	1308	0	0	0	2	28	200	448	547	81	2	0	0
Wind	WSAvg	1.6	1.2	1.4	1.7	1.9	1.8	1.8	1.8	1.7	1.5	1.3	1.3	1.3
Precipitation	PrecAvg	911	62	58	77	83	98	112	83	70	69	70	63	66
	PrecMax	1113	114	101	153	157	178	224	152	129	146	143	118	148
	PrecMin	721	19	36	33	39	45	47	25	20	22	20	17	25
	PrecStd	102	20	17	27	25	29	35	31	26	27	29	31	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	15.9	20.2	25.9	29.9	32.2	34.8	35.0	32.2	25.2	20.0	14.8
		MCWB	6.1	8.3	9.5	15.2	19.4	20.5	23.3	21.5	20.0	15.9	10.7	8.1
	2%	DB	8.0	10.9	16.8	23.0	27.0	30.8	32.8	33.0	29.0	22.2	16.3	10.6
		MCWB	4.1	6.0	9.1	13.7	18.1	20.3	22.0	21.1	18.4	14.9	10.2	6.5
	5%	DB	5.9	7.8	14.0	20.9	25.0	29.0	30.9	31.5	26.8	20.1	13.1	8.0
		MCWB	3.1	4.4	8.1	13.1	17.4	20.2	21.3	20.9	17.9	14.3	8.9	4.8
	10%	DB	3.9	5.1	11.1	18.1	23.1	27.2	29.1	29.9	24.8	18.0	10.8	6.1
		MCWB	1.8	3.1	7.0	11.9	16.6	19.6	20.6	20.7	17.4	13.3	7.8	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	9.1	11.4	16.0	20.7	23.2	25.0	23.6	22.1	17.2	12.3	8.9
		MCDB	10.3	14.7	18.5	23.0	27.6	29.9	32.6	30.4	30.2	22.7	16.6	12.4
	2%	WB	4.6	6.6	9.6	14.5	19.1	21.6	23.2	22.6	19.8	16.0	10.8	7.0
		MCDB	7.2	9.8	14.7	21.5	25.3	28.2	29.8	30.0	25.2	20.6	15.1	9.3
	5%	WB	3.1	4.7	8.6	13.5	18.0	20.7	22.2	21.7	18.8	14.9	9.5	5.6
		MCDB	5.2	7.1	12.9	19.4	23.7	27.0	28.3	29.4	24.7	19.0	12.6	7.6
	10%	WB	2.0	3.3	7.3	12.4	17.0	19.8	21.3	21.1	18.0	13.8	8.2	4.1
		MCDB	3.5	4.9	11.0	17.3	22.3	25.7	27.3	28.2	23.5	17.2	10.4	5.7
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.7	8.0	9.7	9.7	10.0	10.0	10.8	10.0	8.5	7.1	6.5
		MCDBR	9.5	11.2	13.7	14.1	13.1	12.7	12.6	13.4	13.5	12.9	12.5	10.8
		MCWBR	6.7	7.3	7.7	7.0	6.7	5.9	5.3	5.3	5.8	7.1	7.2	7.3
	5% WB	MCDBR	8.7	10.3	12.3	12.3	11.5	11.4	11.2	11.9	11.4	11.0	10.7	9.8
MCWBR		6.4	6.9	7.3	6.6	6.6	6.0	5.5	5.4	5.6	6.6	6.7	6.9	
Clear-Sky Solar Irradiance	taub	0.322	0.358	0.413	0.471	0.442	0.446	0.487	0.484	0.427	0.387	0.342	0.320	
	taud	2.307	2.207	2.108	1.998	2.171	2.212	2.109	2.109	2.239	2.304	2.364	2.326	
	Ebn at Noon	793	818	815	796	835	830	790	776	794	779	764	756	
	Edh at Noon	84	111	139	168	145	140	154	149	121	100	80	76	
All-Sky Solar Radiation	RadAvg	1.59	2.38	3.04	4.27	5.16	5.63	5.70	5.14	3.96	2.63	1.75	1.35	
	RadStd	0.21	0.41	0.32	0.55	0.55	0.36	0.43	0.46	0.44	0.36	0.23	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page